

WASP-50b

R-1782

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_130103 · created 2026-08-02T23:43:34+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456295.6756 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.012
Transit depth (Rp/Rs)^2	2.65 +/- 0.4 [%]
Orbital Inclination (inc)	84.54 +/- 0.91
Airmass coefficient 1 (a1)	1.012 +/- 0.01
Airmass coefficient 2 (a2)	-0.008 +/- 0.0098
Scatter in the residuals of the lightcurve fit is	1.01 %
Best Comparison Star	#3 - [233, 290]
Optimal Aperture	3.62
Optimal Annulus	5.01
Transit Duration (day)	0.0754 +/- 0.008

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.163 ± 0.012 vs 0.1404 (deviation 1.88σ)
Duration fit vs published	0.0754 d vs 0.0754 d (deviation 0.0σ)
Mid-time O-C	-10.2 min
Depth SNR	13.6
Residual scatter	1.01 %

Light curve

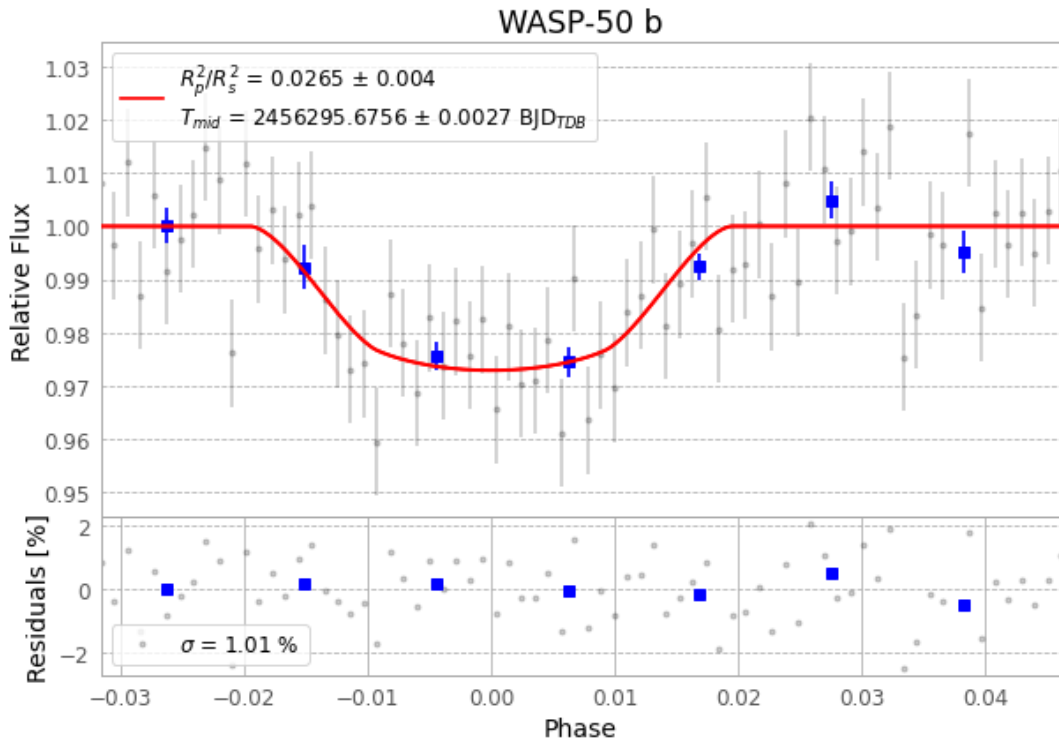


Figure 1 — FinalLightCurve_WASP-50 b_2013-01-02.png (SHA-256 3d941298044c14cf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [245, 313], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7c8bf8d017dcecc4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), WASP-50130103023913.FITS → WASP-50130103061812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-130103.log (290f1bb5836a...), FinalLightCurve_WASP-50 b_2013-01-02.png (3d941298044c...), FinalLightCurve_WASP-50 b_2013-01-02.csv (450a807deff5...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 23:43Z.